

Logarithmic laws



1

a $\log_{10} 20$

b $\log_{10} 6$

c $\log_{10} \left(\frac{1}{4} \right)$

d $\log_5 198$

e $\log_{10} \left(\frac{35}{3} \right)$

f $\log_7 2$

g $3 \log_{10} 18$

h $3 \log_{10} 3$

i $2 \log_5 2$

j $5 \log_{10} 18$

k $\log_4 448$

l $\log_3 (10xy)$

2

a 1

b 3

c 3

d 2

e -2

f 2

3

a $7 \log_4 x$

b $5 \log (x + 6)$

c $-\log (3x + 7)$

d $\frac{2}{5} \log x$

e $5 \log_6 y$

f $\frac{1}{2} \log_6 w$

g $\frac{1}{2}$

h $\frac{5}{4}$

i a

j 10

k $\frac{3}{2} \log_5 x$

l $\frac{15}{4}$

4

a $\log_9 u + \log_9 v$

b $\log_5 9 - \log_5 7$

c $\log_2 5 + \log_2 x$

d $\log_{10} 2 - \log_{10} x$

e $\log m + \log m$

f $5 (\log 3 + \log x)$

g $\log p + \log q - \log r$

h $-\log x - \log y$

i $-7 (\log 5 + \log x)$

j $\log 5 + \frac{2}{3} \log x$

k $\frac{1}{3} (\log 14 + \log x)$

l $4 \log c - \frac{1}{2} \log d$

5

a $7 \log x$

b $\log (x^5 y^3)$

c $\log \left(\frac{x^8}{y^{\frac{1}{3}}} \right)$

d $\log \left(\frac{x^8}{y} \right)$

e $-35 \log_{10} 5$

f $9 \log_{10} 2$

g 2

h 2

6 $\log 2 + \log u - \log 3 - \log v$

7 $5 \log x$

8

a $3 \log u + 5 \log v$

b $\frac{1}{3} \log v - \frac{1}{2} \log u$

9

a 5

b 2



c 9

d -4

e 2

f 3

g 5

h 2

10 2.105

11

a 1.447

b 0.222

c 3.121

d -0.407

12

a 1.24

b 0.31

c 1.78

d -0.62

e 2.02

13

a 7.8

b 1.4

c 7.3

d -1.6

14 $\frac{\log_4 20}{\log_4 3}$

15

a $\frac{\log_{10} 16}{\log_{10} 4}$

b $\frac{\log_{10} 0.9}{\log_{10} 3}$

c $\frac{\log_{10} \sqrt{5}}{\log_{10} 3}$

16

a i $\frac{\log_{10} 21}{\log_{10} 8}$

ii 1.46

b i $\frac{\log_{10} \sqrt{5}}{\log_{10} 4}$

ii 0.58

c i $\frac{\log_{10} 3^3}{\log_{10} 2}$

ii 4.75

d i $\frac{\log_{10} 105}{\log_{10} \pi}$

ii 4.07

17

a $p + q$

b $p - q$

c $2q$

d $2p + 1$